Active Maintenance Report: coppock_green_porter_2022

Coppock, Green and Porter (2022, Research & Politics): Does Digital Advertising Affect Vote Choice?

2026-08-03

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This repository holds the actively maintained replication code for Coppock, Green and Porter (2022), together with the reproducibility report that documents what the original archive did and did not do. It is part of a program applying the maintenance proposal in Peer, Orr and

Coppock (2021, *PS: Political Science & Politics*, doi 10.1017/S1049096521000366) to a set of published archives.

Article	10.1177/20531680221076901
Replication archive	10.7910/DVN/UH2NQW
Pre-analysis plan	egap.org/registration/5312, mirrored at osf.io/ch4ms

The data are not redistributed here. The deposit is 1.9 MB across 14 files and lives at Harvard Dataverse, which is the only copy this repository points at. `download_original.R` fetches it and verifies every file; `original_manifest.csv` pins the file identifiers, sizes and checksums, so the exact bytes this code was written against are recorded in version control even though the bytes themselves are not.

Repository layout. `maintained/` is the maintained rewrite: one script per published table or figure, writing to `output/`, which is committed so a reader can compare a fresh run against it without downloading anything. `ground_truth/` ties every published number to the code that produces it: `published_claims.csv` is the extraction of every numeric token in the article and its appendix, `build_ground_truth.R` assembles the comparison and gates it, and `maintained/in_text_claims.R` recomputes every claimed number a second time by its own path. `original/` is created by the download script and is deliberately absent from the repository. `errata.qmd` renders `coppock_green_porter_2022_errata.pdf`, a short note listing the sentences and floats in the article that are wrong. This file is the reproducibility report, also available as a PDF in `report/`.

License. CC0 1.0 Universal, matching the terms of the deposit this repository maintains. See `LICENSE`.

To reproduce. Clone or download the repository, open `coppock_green_porter_2022.Rproj`, and run:

```
source("run_all.R")
```

That fetches the deposit, verifies its 14 files, produces every table and figure into `maintained/output/`, and then rebuilds and gates the ground truth. Required packages: tidyverse, estimatr, modelsummary, DeclareDesign, knitr, kableExtra, here. Paths resolve through `here`, so nothing depends on the working directory. The full run takes about three minutes, most of it in the two simulation scripts: 2000 re-randomizations for the randomization inference and 5000 design simulations for Figure G.2. A successful run overwrites `maintained/output/`, which is committed: **`git diff` on that folder is the reproduction check.**

1 Summary

Two questions, answered before the detail.

1.1 Does the deposited archive run?

Almost. Eight of the nine analysis scripts execute without error on a current R installation. The ninth, `CGP_2022_equivalence_tests.R`, stops on its first model: it asks `lh_robust()` for a linear combination of coefficients with clustered CR2 standard errors, and current `estimatr` refuses, because `lh_robust()` has no CR2 path. Everything in appendix Table C.6 therefore has no live source in the deposit, which is 30 of the 627 recorded quantities.

Of what does run, every number matches the table it belongs to. Two numbers in the article's prose do not match the table they describe, but the code is right and the sentence is wrong; see the errata. The one class of quantity that cannot reproduce exactly is the simulated one: neither the randomization inference nor the design diagnosis sets a seed, so the archive does not return its own published p-values twice in a row, let alone four years later. Both scripts also ship with the number of simulations cut by a factor of four to ten below the value used for the paper, with the published setting left commented out on the line above.

1.2 Does the maintained rewrite reproduce the paper?

Yes, with the exceptions that are the point of the exercise. 549 of the 613 verifiable ground truth claims match the published values to reported precision. The 64 that do not fall into four groups:

- **52 are cells of appendix Table B.3, which is a reprint of Table B.4.** Every cell beneath a vote share caption is a vote margin cell, so the rewrite's vote share models disagree with all of them. The four that do agree are the cluster counts, which a table printed in the wrong place still gets right. See the errata.
- **Two are errors in the published text.** Describing Table 4's first column, the article says the unadjusted estimate is "a 2.1 percentage point increase" with "a standard error of 3.0 percentage points". Table 4 gives 0.0021 with a standard error of 0.0225, which is 0.21 and 2.25 percentage points. The estimate in the text is off by a factor of ten and the standard error does not correspond to any model in the paper, and it inflates the reported effect tenfold in the only place a reader meets it in prose.
- **Seven are simulation draws.** The two randomization inference p-values, the two power figures as the appendix states them, the two as Figure G.2 prints them, and the posterior power as the main text states it. The published values and the rewrite's are two independent draws, and every pair differs by less than two standard errors of that difference; the largest gap, 0.508 against 0.534 for the adjusted randomization inference p-value, is 1.7. They cannot be matched exactly, because the archive sets no

seed. Rounded to whole per cent, as the main text states them, the OLS power matches at 21 and the posterior power of 89.9 rounds to 90 against the text's 89.

- **Three are figure labels.** Figure G.2's two panels annotate the simulated effect as `PATE = 0.1` where the design draws from a normal centred on 0.01, and the published Figure 1 annotates this study's standard error as 0.009 because the archive typed the table's rounded 0.0085 into the figure. The rewrite reads the estimate from the Table 4 output instead, and 0.0084760 rounds to 0.008. Nothing plotted in either figure is affected.

The remaining 6 recorded quantities are unverifiable rather than unmatched: four are precinct counts for a sample the deposit does not contain, and two are the range in "approximately 10 to 20 voting precincts", which describes the same absent mapping.

2 Paper overview

Citation: Coppock, A., Green, D. P. and Porter, E. (2022). "Does digital advertising affect vote choice? Evidence from a randomized field experiment." *Research & Politics*, 9(1). DOI: 10.1177/20531680221076901

Summary: A blocked, clustered field experiment run during the 2018 midterms tests whether pre-roll video advertisements on Facebook and Instagram move vote choice. Two hundred and ten Florida ZIP codes across four congressional districts were grouped into matched trios; one ZIP in each trio was assigned to treatment and then to one of two anti-Republican advertisements on gun policy, neither of which named a candidate. The advertisements ran from 25 October 2018 until each exhausted a $30,000 budget, generating 1.1 million three-second views. The outcome is the precinct-level Democratic two-party vote share from the Florida Secretary of State, analysed on the 853 precincts that fall wholly inside a study ZIP code and have outcome data, with CR2 standard errors clustered by ZIP code and one-tailed randomization inference under the sharp null. The unadjusted estimate is 0.21 percentage points (SE 2.25); adjusting for the 2012, 2014 and 2016 vote shares brings it to -0.04 points (SE 0.85). A Bayesian integration of this result with the three prior field experiments on digital advertising leaves a posterior centred on 0.7 percentage points with a standard deviation of 0.4.

3 Original archive reproducibility

Archive: Harvard Dataverse, 10.7910/DVN/UH2NQW, published 5 January 2022. Fourteen files inside a `replication_archive` directory: nine analysis scripts, two `.rds` data files, a README, a session log, and an `.Rapp.history` file carrying one line of a coauthor's local path. The README names `tidyverse`, `DeclareDesign`, `xtable` and `texreg` as the required add-ons and records R 4.1.1 under macOS Big Sur.

Table 1: Original archive, run script by script in a

Script	Status
CGP_2022_in_text_calculations.R	Clean
CGP_2022_precinct_level_analysis.R	Clean
CGP_2022_zip_code_level_analysis.R	Clean
CGP_2022_precinct_level_analysis_CD.R	Clean
CGP_2022_design_tests.R	Clean
CGP_2022_bayesian_learning.R	Clean
CGP_2022_precinct_level_randomization_inference.R	Runs, but unseeded and at a quarter of the publish
CGP_2022_design_diagnosis.R	Runs, but unseeded and at a tenth of the published
CGP_2022_equivalence_tests.R	Fails on the first model

Classification: resolvable. One script fails and the fix is arithmetic rather than a change of method.

The failure. `lh_robust()` fits the model and then evaluates a linear hypothesis on the fitted coefficients. Current `estimatr` will not do this with CR2 standard errors, which are what `clusters =` implies, so all six of the script's models stop with `lh_robust not available for CR2 standard errors`. The quantity wanted is a difference between two coefficients in a model that `lm_robust()` fits without complaint, so the rewrite takes the difference and its standard error from the fitted coefficient vector and CR2 variance-covariance matrix directly. That is the same arithmetic `lh_robust()` performed, and it returns Table C.6 exactly.

Two claims in the archive that turn out to be false, or at least stale. Both simulation scripts carry a line of the form `# simulations <- simulate_design(design, sims = 2000)` immediately above the line that runs with `sims = 500`. The implication is that the published numbers came from the larger run and that the smaller one is there for the replicator's convenience. Measured, the larger runs cost 32 seconds and 55 seconds on a 2026 laptop, so there is no convenience to buy. The rewrite runs at the published simulation counts.

Style-level issues, none fatal. `rm(list = ls())` opens all nine scripts. `%>%` throughout. `do(tibble(...))` in the Bayesian learning script, superseded by `reframe()`. `texreg()` and `print.xtable()` write LaTeX to standard output, so the tables in the paper were produced by copying console output rather than by writing a file. Every path in the scripts is bare, with a `# setwd() to replication archive` comment standing in for the working directory.

One deposited file is not part of the analysis. `.Rapp.history` contains a single `load()` call pointing at `/Users/eporter/Dropbox/...` and a file, `CGP_2022_precinct_level_ep_rep.rds`, that is not in the deposit. It is a stray console history, harmless but worth noticing, since it is the only trace of the working directory the analysis was actually run in.

4 Errata

18 defects in the published record, none of which changes a substantive conclusion. They are published as a standalone note, `coppock_green_porter_2022_errata.pdf`, rendered from `errata.qmd`, which is the single definition of every entry: the note, this section and the remastered edition all read the same `errata_entries.csv` rather than keeping copies, because when they kept copies they diverged.

The 7 below are the ones the rewrite corrects or flags, numbered as the note numbers them. The remaining 11 are errors in the reference list, nearly all introduced by the publisher's reference processing, found by an audit that checks every printed entry against Crossref; they are set out in the note and do not bear on the reproduction.

1. The text misstates the unadjusted estimate by a factor of ten. "The first column shows the unadjusted difference-in-means estimate of the treatment effect of any video: a 2.1 percentage point increase in Democratic vote share, though this estimate is uncertain, as evidenced by its large standard error of 3.0 percentage points." Table 4 reports 0.0021 (0.0225). Converted to percentage points, that is 0.21 with a standard error of 2.25. The estimate is stated at ten times its value, and the standard error matches no model in the paper or appendix. The sentence's conclusion is unaffected, since both readings describe an estimate that is small and swamped by its standard error, but the number itself is wrong.

2. Figure G.2's panel annotation names the wrong effect size. The label reads "Power when PATE = 0.1". The simulated effects are drawn from a normal centred at 0.01, which is also what the appendix text says two paragraphs above. The string is hardcoded in the archive script. The rewrite prints 0.01.

3. Covariate row labels are wrong in five appendix tables. Thirteen covariate rows name a variable their model does not contain. Tables E.7 and F.8 regress the treatment indicator on lagged two-party *vote margins* and label all three lagged rows "Two Party Vote Share". Tables F.9, F.10 and F.11 pass `texreg` a coefficient-name vector whose first covariate entry is "Missingness Indicator (2016)"; no ZIP code is missing 2016 returns, so the ZIP-level models carry no such indicator and the label lands on the 2016 vote share, margin or total coefficient instead. The same vector is reused for all three outcomes, which is why the 2014 and 2012 rows of F.10 and F.11 read "Two Party Vote Share" over vote margins and vote totals. The 2014 and 2012 rows of F.9 are correct as published. No coefficient moves; only its name does. The rewrite labels each row with the variable actually in the model.

4. Appendix Table B.3 prints Table B.4. The caption of B.3 reads "Effects on vote share (CD fixed effects)", and every cell beneath it is identical to Table B.4, "Effects on vote margin (CD fixed effects)": the same coefficients, standard errors, R-squared values and covariate labels, all on the vote margin scale. The vote share models with congressional district fixed effects are in the archive but appear nowhere in the appendix. The rewrite computes them; `maintained/output/table_b3_vote_share_precinct_cd.csv` is where they live. The

estimate on any treatment video is 0.0042 (0.0217) unadjusted and -0.0018 (0.0088) adjusted, which is the same substantive story as Table 4.

5 Ground truth

Table 2: Ground truth, by published object

Table or figure	Claims	Archive verifiable	Archive matches	Rewrite verifiable
Abstract	2	2	2	2
Appendix A	1	0	0	0
Appendix C	3	0	0	0
Appendix D	1	0	0	1
Appendix E	2	1	1	1
Appendix F	1	0	0	1
Appendix F.1	2	1	1	1
Appendix G	4	2	0	4
Appendix H (pre-analysis plan)	4	0	0	4
Bayesian integration	9	6	5	8
Discussion	3	2	2	2
Field experiment: Florida advertisements	7	0	0	5
Figure 1	12	12	12	12
Figure G.2	4	2	0	4
Note 1	4	4	4	4
Results	7	6	2	7
Table 1	24	20	20	20
Table 4	56	20	20	56
Table A.1	56	20	20	56
Table A.2	56	16	16	56
Table B.3	56	12	0	56
Table B.4	56	16	16	56
Table B.5	56	16	16	56
Table C.6	30	0	0	30
Table E.7	17	5	5	17
Table F.10	48	16	16	48
Table F.11	48	16	16	48
Table F.8	10	4	4	10
Table F.9	48	16	16	48

Every value in the `value_paper` column of `ground_truth/coppock_green_porter_2022_ground_truth.csv` was read off the published article or its appendix, and it is stored as the string the article prints rather than as a number, because a double does not record how many decimals were on the page. A value agrees when the rewrite's number, printed to that precision, gives the same digits.

`value_script` is what the deposited scripts produce. It covers 215 of the 627 rows: those are the cells measured from a run of the deposit and recorded in `ground_truth/archive_values.csv`. This repository has no `extract_archive_values.R`, so the archive column is a recorded measurement rather than a generated one, and the rows it does not reach carry `match = NA` rather than a verdict.

Table 3: The 12 claims outside Table B.3 the rewrite does not match

Claim	Published
Appendix G: Power of the Bayesian posterior estimator, per cent, as stated in the appendix	89.3
Appendix G: Power of the study on its own, per cent, as stated in the appendix	21.0
Figure 1: The present study's standard error on the panel face	0.009
Figure G.2: PATE named in the left panel annotation	0.1
Figure G.2: Power printed in the left panel	0.210
Figure G.2: PATE named in the right panel annotation	0.1
Figure G.2: Power printed in the right panel	0.893
Bayesian integration: Power of the experiment on its own, per cent, as stated in the main text	21
Results: One-tailed randomization inference p-value, covariate-adjusted model	0.508
Results: One-tailed randomization inference p-value, unadjusted model	0.471
Results: Unadjusted effect on Democratic vote share as stated in the text, percentage points	2.1
Results: Standard error of the unadjusted effect as stated in the text, percentage points	3.0

`defect_locus` records where the fault lies, because a mismatch otherwise reads as a failure of the rewrite and here never is. `paper_internal` means the article disagrees with its own tables, `archive` that the deposit cannot support the claim, and `environment` that a package moved underneath the deposited code.

Table 4: Where the fault lies, by row

Defect locus	Rows
archive	16
environment	30
paper_internal	54

6 The extraction and the two instruments

The ground truth answers "does this number reproduce?" only for numbers somebody wrote down. What decides whether the answer means anything is the extraction: `ground_truth/published_claims.csv` lists **every numeric token in the article and its online appendix**, 754 of them, each classified by hand and each carrying the string the page prints. Spelled-out numbers were swept for separately, which is where "three conditions", "matched trios" and "four possible explanations" come from. Bibliographic apparatus is excluded: citation years, the reference list, the DOI, page headers and affiliation markers.

Table 5: The extraction, by claim type

Claim type	Needs a block	Verified at the point of use
definitional	5	27
descriptive	10	0
pipeline	612	0
structural	0	62
transcribed	0	38

`pipeline` claims are estimates. `descriptive` claims are assertions about shape, sign or count with no printed number, such as "each of the individual coefficients is nonsignificant"; each gets a computed truth value in the `holds` column. `transcribed` claims are other people's results and the advertisement vendor's exposure counts, which cannot drift with this pipeline. `definitional` and `structural` claims are design parameters, cross-references and front matter; they get a block only where the pipeline can actually reach them, which here means the prior's standard deviation and the two parameters of the simulated effect distribution.

Coverage, float by float. A float covered by one row is a float nobody has checked, so the table below states the covered fraction for each published object rather than reporting that each has at least one row.

Table 6: Published numbers, and how many of them are checked

Published object	On the page	Ground truth rows	Verified
Abstract	5	2	2
Appendix A	1	1	1
Appendix C	5	3	3
Appendix contents	10	0	0
Appendix D	2	1	1
Appendix E	2	2	2
Appendix F	3	1	1

Appendix F.1	2	2	2
Appendix front matter	2	0	0
Appendix G	4	4	2
Appendix H (pre-analysis plan)	28	4	4
Bayesian integration	14	9	8
Discussion	5	3	3
Field experiment: Florida advertisements	26	7	5
Figure 1	21	12	11
Figure D.1	12	0	0
Figure G.2	7	4	0
Introduction	1	0	0
Note 1	7	4	4
Results	11	7	3
Table 1	24	24	20
Table 2	5	0	0
Table 3	6	0	0
Table 4	56	56	56
Table A.1	56	56	56
Table A.2	56	56	56
Table B.3	56	56	4
Table B.4	56	56	56
Table B.5	56	56	56
Table C.6	30	30	30
Table E.7	17	17	17
Table F.10	48	48	48
Table F.11	48	48	48
Table F.8	10	10	10
Table F.9	48	48	48
The challenge of political persuasion	14	0	0

Two instruments, not one. `ground_truth/build_ground_truth.R` and `maintained/in_text_claims.R` reach the same claimed numbers from the same files in `maintained/output/` by separate paths, each doing its own selection, unit conversion and rounding. The build pivots each output file long and matches coefficient keys; the claims file filters row by row and prints each value beside the sentence that states it. Neither refits: estimation happens once, in the analysis scripts, and only derivation happens twice.

`in_text_claims.R` prints one line per claim, `CLAIM <id> = <value> || <label>`, and the printed id is the only link between a block and the claim it covers. `build_ground_truth.R` runs that file as a program, captures what it printed, and stops unless:

- every extraction row that needs a block printed one, and every printed id names a claim the extraction declares;
- the number of printed claims equals the number of rows requiring one, 627 either way;
- every `pipeline` and `descriptive` claim has a ground truth row;
- the two instruments agree value by value, at the precision the article printed;
- every row carrying an adverse verdict, or one the deposit cannot answer, names a `defect_locus`, and every clean match names none;
- every `value_paper` string round-trips through its own precision, which catches a transcription whose digits are right and whose precision is not.

The gate matters because the coverage boundary is where drift accumulates. The most consequential error in this article is not in any float: the tenfold overstatement on page 4 is prose, and a ground truth built from the pipeline outward would never have posed the question.

7 Maintained rewrite

Table 7: Maintained rewrite: one script per published object

Script	Produces
helpers.R	Packages, the cluster-count glance method, the table footer spec
table_1_sample_description.R	Table 1
table_4_vote_share_precinct.R	Table 4
table_a1_vote_margin_precinct.R	Table A.1
table_a2_turnout_precinct.R	Table A.2
table_b3_vote_share_precinct_cd.R	Table B.3, as it should have appeared
table_b4_vote_margin_precinct_cd.R	Table B.4
table_b5_turnout_precinct_cd.R	Table B.5
table_c6_equivalence_tests.R	Table C.6
table_e7_balance_precinct.R	Table E.7 and its joint F test
table_f8_balance_zip.R	Table F.8 and its joint F test
table_f9_vote_share_zip.R	Table F.9
table_f10_vote_margin_zip.R	Table F.10
table_f11_turnout_zip.R	Table F.11
figure_1_bayesian_learning.R	Figure 1
figure_g2_design_diagnosis.R	Figure G.2
text_in_text_calculations.R	The vote totals, posteriors and cost per vote quoted in the text
text_randomization_inference.R	The two randomization inference p-values quoted in the text
in_text_claims.R	Every published number, recomputed and printed beside its senten

Every script writes a full-precision `.csv` of every coefficient it estimates, a `_gof.csv` of the R-squared, sample size and cluster count that sit at the foot of the published table, and a `.tex` rendering with the same rows and columns as the published table, so a reader can compare them line by line. The summary rows are written out rather than read back off the formatted table, for the same reason the coefficients are: reading a rounded cell and rounding it again manufactures mismatches.

Substitutions.

Archive	Rewrite	Why
`lh_robust()` with clusters	`lm_robust()` plus the contrast taken from the CR2 variance-covariance matrix	`lh_robust()` has no CR2 path; the arithmetic is identical
`texreg()` and `print.xtable()` to standard output	`modelsummary(output = )`	The table is written to a file rather than copied out of a console
`do(tibble(...))`	`reframe()`	`do()` is superseded
`DeclareDesign:::format_num()`	`sprintf()`	Unexported internals are not a stable interface
`separate()`	`separate_wider_delim()`	`separate()` is superseded
`%>%`, `ifelse()`, `rm(list = ls())`	`|>`, `if_else()`, omitted	House style
`sims = 500`, no seed	`sims = 2000` and `sims = 5000`, seeded	The published simulation counts, at a measured cost of 32 and 55 seconds

No published number is an input to a computation. The archive's Figure 1 script typed this paper's own estimate and standard error into the vector of experiments being pooled, and the in-text script typed the vote total and the posterior mean into the cost-per-vote formula. In the rewrite, `figure_1_bayesian_learning.R` reads the estimate out of `output/table_4_vote_share_precinct.csv`, and `text_in_text_calculations.R` reads the posterior out of `output/figure_1_bayesian_learning.csv` and the vote total out of the ZIP-level data. The only constants entered by hand are the three prior studies' published estimates, which are other people's results and belong in the code, and the campaign's budget, which is a design fact rather than an estimate.

The cost per vote depends on which posterior you use. The article's footnote computes $100{,}000/(822{,}783 \times 0.007 \times 2) = \$8.68$ from the rounded quantities it quotes. The unrounded posterior is 0.00708, and the same calculation at full precision gives \$8.58. The rewrite reports both, and the ground truth compares the first, since that is the number in the paper.

8 Figure verification

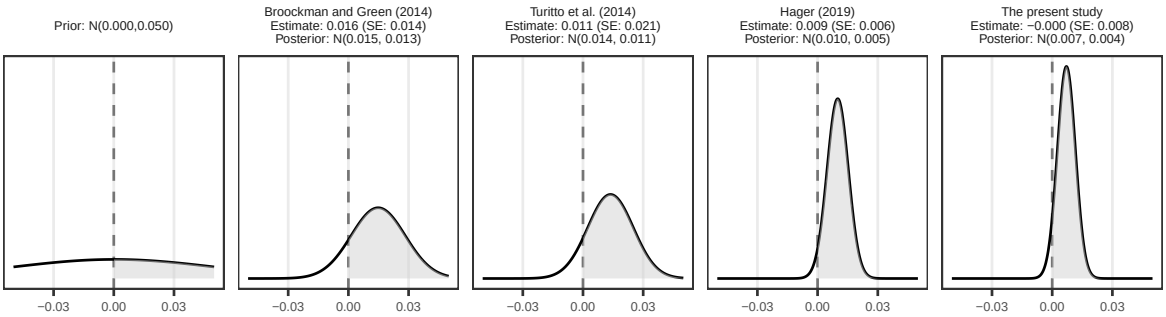


Figure 1: Figure 1, maintained rewrite. Compare against the published figure at doi 10.1177/20531680221076901.

Table 8: Figure 1, panel by panel

Panel	Estimate entering the update	Posterior mean	Posterior SD
Prior		0.0000	0.0500
Broockman and Green (2014)	0.0160	0.0148	0.0135
Turitto et al. (2014)	0.0110	0.0137	0.0113
Hager (2019)	0.0090	0.0100	0.0053
The present study	-0.0004	0.0071	0.0045

Every posterior matches the published panel labels: N(0.015, 0.013), N(0.014, 0.011), N(0.010, 0.005) and N(0.007, 0.004). The one visible difference is the standard error annotated on the last panel, which is 0.008 here and 0.009 in the article, for the reason given in the summary.

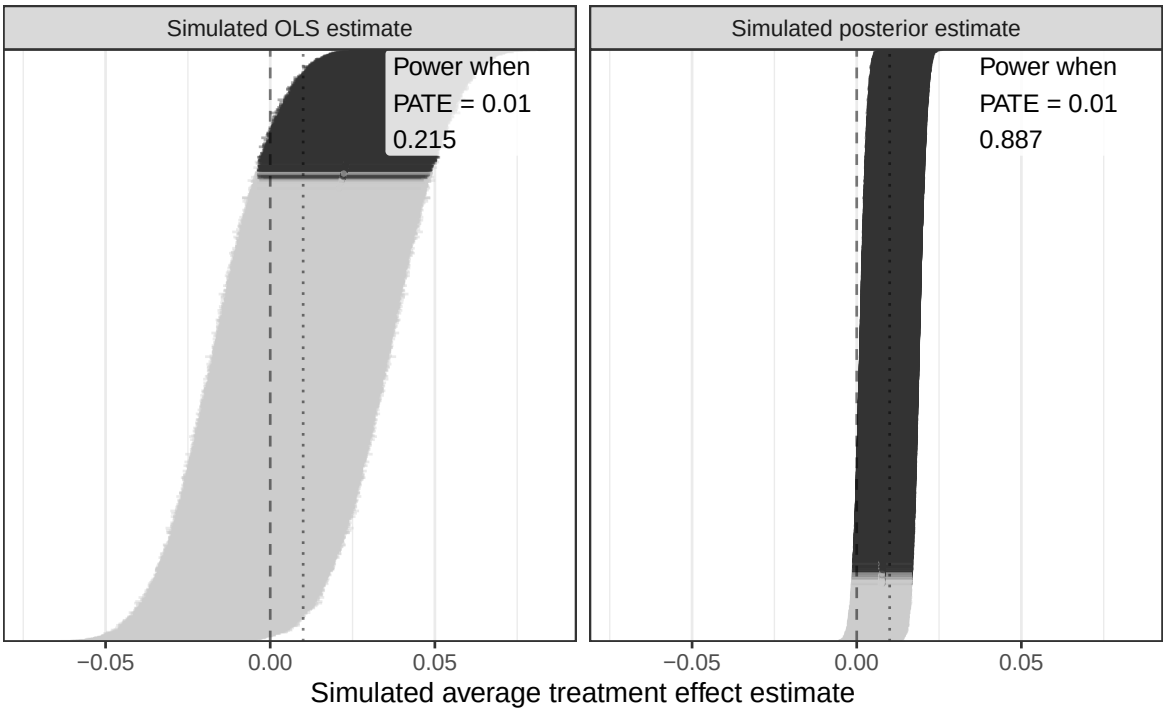


Figure 2: Figure G.2, maintained rewrite, at the published 5000 simulations.

Table 9: Figure G.2 diagnosands

Simulations	Mean OLS estimate	Power, OLS	Power, posterior
5000	0.0098	0.215	0.887

The appendix reports power of 0.210 for the OLS estimator and 0.893 for the posterior. Treating the published figures and these as two independent draws of 5000, the gaps are 0.3 and 1.0 standard errors of their difference, which is as close as an unseeded simulation permits.

9 Rewrite verification

Table 10: Rewrite against the published values

Table or figure	Claims	Matching	Not matching
Abstract	2	2	0
Appendix D	1	1	0

Appendix E	1	1	0
Appendix F	1	1	0
Appendix F.1	1	1	0
Appendix G	4	2	2
Appendix H (pre-analysis plan)	4	4	0
Bayesian integration	8	7	1
Discussion	2	2	0
Field experiment: Florida advertisements	5	5	0
Figure 1	12	11	1
Figure G.2	4	0	4
Note 1	4	4	0
Results	7	3	4
Table 1	20	20	0
Table 4	56	56	0
Table A.1	56	56	0
Table A.2	56	56	0
Table B.3	56	4	52
Table B.4	56	56	0
Table B.5	56	56	0
Table C.6	30	30	0
Table E.7	17	17	0
Table F.10	48	48	0
Table F.11	48	48	0
Table F.8	10	10	0
Table F.9	48	48	0

Every printed cell of every published table is a row in the table above, so "reproduces the published table" is a counted claim here rather than an impression: Table 4, Tables A.1 and A.2, Tables B.4 and B.5, Tables C.6, E.7 and F.8, and Tables F.9 through F.11 match in all 501 of their cells. Table B.3 is the exception, and it is the erratum.

Determinism. Every script other than the two simulation scripts is deterministic and returns byte-identical output on a second run. The two simulation scripts are seeded, so they are also reproducible run to run; they are not reproducible against the archive, which set no seed.

10 R environment

Table 11: Versions this report was built against

Component	Version
R	4.6.1
tidyverse	2.0.0
estimatr	2.0.1.9000
modelsummary	2.6.0
DeclareDesign	2.0.0
knitr	1.51
kableExtra	1.4.0
here	1.0.2

The archive was written against R 4.1.1 with `estimatr` 0.30.2 and `DeclareDesign` 0.28.0. The `lh_robust()` failure is the only place where the four intervening years of package development broke something outright.